



Discrete Structures I

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Logic and Proofs

- This chapter will set the foundations of mathematical reasoning and thinking to be used in this course and in all your subsequent math and computer science courses.
- Fundamental to all mathematics and computer science is logical arguments and proofs.
- In this first chapter we will explore different logic systems, namely propositional logic and predicate logic. Within these systems we will see how to form formulas, logical arguments, inferences, and how to prove with certainty if some mathematical statement is correct or not.
- The reason we need to study logic more formally is that English, and many natural languages, are very ambiguous.



Logic and Proofs

- Propositional Logic is the logical system built around propositions. From such propositions one can build logical arguments and implications.
- In this section we will explore the language of propositions, their applications, and deriving logical equivalences.



Logic and Proofs

- Propositions are all about truth. Is a statement true or false? Is a statement correct or incorrect?
- A proposition is a declarative sentence or statement has a truth value. It is a statement which is either true or false. Each proposition has a "truth value": either true or false.
- Examples of propositions
 - Socrates was a human.
 - Socrates was a pigeon.
 - It is raining today.
 - The logo of Starbucks is a green mermaid.
 - $1 + 1 = 2$
 - $1 + 1 = 4$



Logic and Proofs

- Notice that propositions need not be a true statement. Propositions only need to be declarative. Their truth value may be true or false.
- However, all propositions must have a particular truth value. The statement cannot be both true and false. The statement must be able to be interpreted as true or false.
- From the previous definition and examples, propositions are therefore not questions, general statements, demands, or hypotheses. Propositions do not have any variables, quantifiers, or parameters (e.g. the words “some” or “any” typically do not appear). Consider now a few non-examples.



Logic and Proofs

- Examples of statements that are not propositions
 - Do you have a dog?
 - Let's go!
 - Some coffee mug with a mermaid on it.
 - $x + 2 = 3$
 - $y = x^2 - 1$
- Are each of these propositions?
 - I am a Quail.
 - Jupiter is the 5th planet from the sun.
 - On Thursdays, van Gogh painted landscapes.
 - $\frac{11+56*3-8}{19} = 9$



Logic and Proofs

- An entire proposition is often denoted by a single propositional variable.
- Propositional variables are typically among $p, q, r, s, t \dots$
- Using propositional variables
 - $p :=$ The sky is blue
 - $q :=$ The sun rises from the west
- Denote truth values in particular ways. "True" may be denoted by T
- "False" may be denoted by F



Logic and Proofs

- Propositions (and propositional variables) can be combined into compound propositions or propositional formulas. Similar to compound sentences and other logical connectives in natural language.
- Propositional Logic has 5 main connectives. Each connective has a corresponding meaning in natural language
 - 1 Negation: $\neg$
 - 2 Conjunction: $\wedge$
 - 3 Disjunction: $\vee$
 - 4 Implication: $\implies$
 - 5 Biconditional $\iff$
- Logical connectives are like arithmetic operators $+$, $-$, $\times$, $\div$



Negation

- The negation of a proposition results in a proposition with the opposite truth value
- Given a proposition p its negation is $\neg p$ and has the following truth values.

p	$\neg p$
T	F
F	T

- Let $p =$ "the sky is blue". What is $\neg p$?
- Let $q = 2 + 2 = 5$. What is $\neg q$?



Conjunction

- The conjunction of two propositions is the logical "and" of the two propositions.
- The conjunction of two proposition is only true if both the propositions are individually true, otherwise the conjunction is false.
- Given proposition p and q their conjunction is denoted $p \wedge q$ and has the following truth values.

p	q	$p \wedge q$
F	T	F
F	F	F
T	T	T
T	F	F



Disjunction

- The disjunction of two propositions is the "or" of the two propositions.
- The disjunction is true if at least one of the propositions is individually true, otherwise the disjunction is false.
- Given proposition p and q their conjunction is denoted $p \vee q$ and has the following truth values.

p	q	$p \vee q$
F	T	T
F	F	F
T	T	T
T	F	T



Implication

- Implication is one of the most challenging connectives to understand.
- Yet it is arguably the most important for creating logical arguments
- An implication is a *conditional statement*. For two propositions p and q , $p \implies q$ is an implication which is read "if p , then q ". You can also say " p implies q ".
- Let p be "it is raining" and q be "the ground is wet", what is $p \implies q$?
- Let t be "I score an A in KCSE" and s be "I will graduate with a First", what is $t \implies s$?
- In an implication $p \implies q$, the first proposition p is known as the *hypothesis*, *antecedent*, or *premise*. The second proposition q is known as the *conclusion* or *consequence*.



Implication

- Since an implication is a conditional, the truth value of the implication as a whole changes depending on the truth value of the premise.
- The following truth table summarizes the truth values of an implication.

p	q	$p \implies q$
F	F	T
F	T	T
T	F	F
T	T	T

- An implication can be viewed as an *obligation*, a *contract*, or a *commitment*.
- The implication $p \implies q$ is false (the contract is broken; the obligation is unmet) only when p is true and q is false.



Implication

- If p is true, then $p \implies q$ is always true.
- If p is true and the implication correct (the obligation is upheld), then q can never be false.
- "Falsity can imply anything." If the hypothesis is false, then the implication is always true, regardless of the whether or not the conclusion is true.



Implication

- There are many equivalent ways to think about the implication

$$p \implies q$$

if p , then q	q whenever p
$p \implies q$	q follows p
q when p	p is sufficient for q
q , if p	q is necessary for p



Necessity and Sufficiency

- An implication connects propositions by a necessary or sufficient condition. From $p \implies q$ we get two relations:
 - ① p is sufficient for q
 - ② q is necessary for p
- That is, "if *sufficient* condition, then *necessary* condition".



Necessity and Sufficiency

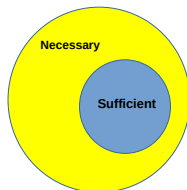


Figure: Being in the inner circle is sufficient for being in the outer circle. Being in the outer circle is necessary for being in the inner circle.



Biconditional

- For two propositions p and q , they can be connected by a biconditional as $p \iff q$.
- A biconditional is an double implication. A biconditional is true if both propositions have the same truth value. $p \iff q$ can be read as "if and only if". A biconditional has the following truth table.

p	q	$p \iff q$
F	F	T
F	T	F
T	F	F
T	T	T



Biconditional

- "p if and only if q"
- "if p then q, and if q then p"
- "p is necessary and sufficient for q"
- "p iff q"

What is the truth value of these compound propositions?

- "The Earth is flat" $\implies$ "Pigeons are robots"
- "Bats have wings" $\implies$ "Bats are birds"
- "A square is a rectangle" $\iff$ "A square had four 90° interior angles"
- "Spinach is green" $\iff$ "Penguins can fly"



Propositional Formulas

- Much like arithmetic formulas using addition, multiplication, division, etc., propositional formulas may use several connectives simultaneously.
- Logical connectives have a similar order of precedence to BEDMAS.
 - 1 Parenthesis: always perform operations on expressions inside parentheses first.
 - 2 Negation: apply negation to a proposition before binary connectives.
 - 3 Conjunction: conjunction before disjunction
 - 4 Disjunction: disjunction after conjunction, but before implication
 - 5 Implication: $\implies$ after $\wedge, \vee$,
 - 6 Biconditional: $\iff$ after $\wedge, \vee$ and $\implies$.
- Propositional variables need not be associated with a particular proposition or truth value.
- Replacing the variables in a propositional formula with a truth value is called a truth assignment.



Propositional Formulas

Definition 1.1: Truth Assignment

A truth assignment is the assignment of a truth value (true or false) to a propositional variable. Equally, it is the replacement of a propositional variable with a truth value.

- Propositional formulas result in different truth values depending on the particular truth assignment on its constituent propositional variables.
- When at least one truth assignment exists so that a formula is true, that formula is said to be satisfiable.

Definition 1.2: Satisfiable

A propositional formula is satisfiable if its truth value can be true under some truth assignment. If every possible truth assignment makes the formula have false as its truth value, that formula is said to be unsatisfiable.



Truth Tables

Truth tables are tools for determining the truth values of propositional formulas

- Table is separated into two sets of columns
 - ① The first set of columns represent each proposition (or propositional variable) in a formula
 - ② The second set of columns represents the sub-formulas and formulas whose truth value are to be determined
- There must be one row in the table for every possible combination of truth values of the propositional variables.



Truth Tables - 3 Variable Truth Table

Let p, q, r be propositional variables. A truth table for the formula $(p \wedge q) \vee r$ is

p	q	r	$p \wedge q$	$(p \wedge q) \vee r$
F	F	F	F	F
F	F	T	F	T
F	T	F	F	F
F	T	T	F	T
T	F	F	F	F
T	F	T	F	T
T	T	F	T	T
T	T	T	T	T

p, q, r have every possible combination of truth values



Truth Tables

- Begin Truth table construction by filling out the columns corresponding to each propositional variable.
- These columns represent every possible combination of truth values on these variables.
- Then add columns for each sub-formula, one at a time, building up to the final formula
- Consider the formula $p \wedge q \wedge r \vee \neg q \vee r \implies p$. Then by order of precedence this equation becomes $((p \wedge q \wedge r) \vee ((\neg q) \wedge r)) \implies p$.
- This contains several sub-formulas which we can parse as follows
 - i $\neg q$
 - ii $\neg q \wedge r$
 - iii $p \wedge q$
 - iv $(p \wedge q) \wedge r$
 - v $(p \wedge q \wedge r) \vee (\neg q \wedge r)$
 - vi $((p \wedge q \wedge r) \vee (\neg q \vee r)) \implies p$



Truth Tables

- To be explicit as possible, create a truth table with $3 + 6 = 9$ columns (3 variables, 6 sub-formulas).
- A truth table for the propositional formula $p \wedge q \wedge r \vee \neg q \wedge r \implies p$

p	q	r	$p \wedge q \wedge r$	$\neg q \wedge r$	$(p \wedge q \wedge r) \vee (\neg q \wedge r)$
F	F	F	F	F	F
F	F	T	F	T	T
F	T	F	F	F	F
F	T	T	F	F	F
T	F	F	F	F	F
T	F	T	F	T	T
T	T	F	F	F	F
T	T	T	T	F	T



Implication, inverse, converse and contrapositive

- Consider the propositional formula $p \implies q$

① **Converse:** $q \implies p$

② **Inverse:** $\neg p \implies \neg q$

③ **Contrapositive:** $\neg q \implies \neg p$

Definition 1.3: A conditional and its inverse

The proposition "if it is raining, then I wear a jacket" is a conditional statement. Its inverse is "if it is not raining I do not wear a jacket"

- Note** An implication and its inverse are not exactly the same.



Applications of Propositional Logic

- Generally propositional logic is applied in Logic, Computer Architecture, Theoretical computer science among others
- More formally, PL is used in:
 - a Translating English to Logic
 - b Boolean Searches
 - c Formal specifications of software and computer systems
 - d Solving puzzles



Translating English to Logic

- Translating English sentence to propositional formula is a 2-Step activity
 - ① Find the atomic propositions - The smallest clauses of the sentence which do not contain connectives and represent them using a unique variable
 - ② Determine the appropriate logical connectives for those propositions

Example 1.1

"If it is raining and I am going outside, I bring an Umbrella".
Let p be "it is raining", q , "I am going outside", r "I bring an umbrella". Choosing the connectives right converts the above sentence to $p \wedge q \implies r$

- Do the same for "The Dog is big and friendly or small and tough"



Boolean Searches

- Boolean refers to a special kind of mathematics that deal with truth values: true and false.
- Boolean algebra is a set of rules for manipulating formulas containing variables and truth values.
- Very similar to but distinct from propositional logic.
- **Boolean Searches** is about using logical connectives to help search through datasets and filter pieces of data. Useful in databases and internet search engines.

Example 1.2

Consider keywords Data and Science. **AND** search for records containing both *Data and Science*. **OR** Searches for records containing *Data or Science*, **NOT** Searches records that excludes both keywords.

Web searches is an extension of the same concept



Formal Specifications

- In software, computer and electrical engineering, the requirements of a system software or device must meet precise specifications
- Naturally natural language are often ambiguous and for system requirement to avoid contradiction, those requirements must be translated into precise logical statements.

Example 1.3: Logical Specification

Mute all notifications during business hours when the user's status is not available. Let p be "It is during business hours", q be "the user's status is set to be available" and r be "mute all notifications". A propositional formula for this specification is:

$$p \wedge \neg q \implies r$$

- Propositional Logic can be used to determine if the collection of all requirements for a system is consistent



Formal Specifications

Definition 1.4: Consistent

A set of propositional formulas is consistent if there exists at least one truth assignment so that every proposition is simultaneously true.

- Consistency is not the same as saying every formula in a set is being satisfiable.
- In this case formulas must be simultaneously satisfied by the same truth assignment.
- To determine consistency, we move as follows. Given a collection of propositions $p_1, p_2, \dots, p_n$, then the propositions are consistent if their conjunction is satisfiable.
- The following formula has at least one truth assignment that makes it true $p_1 \wedge p_2 \wedge p_3 \wedge \dots \wedge p_n$
- To determine the consistency of a requirement set, we must first build propositional formulas for each requirement.



Formal Specifications

- If the same clause exists in multiple requirements, the same propositional variable is used.
- Next we assign the same true or false value to each variable so that every formula is simultaneously true.
- Where no assignment exists, the set of requirements are said to be inconsistent.

Example 1.4

Consider an electronic messaging network requiring that all sent messages are certain to be received. This specification behaviour might be modelled as the follows:

- ① Messages remain in the outbound message queue until they have been received by the recipient
- ② An unreceived message is either a draft or is in the outbound message queue
- ③ Received messages cannot be drafts



Formal Specifications

- Let p be "messages is in outbound message queue", q be "Message is received" and r be "message is a draft". Then we have

- $p \implies \neg q$

- $\neg q \implies (p \vee r)$

- $q \implies \neg r$

p	q	r	$p \implies \neg q$	$\neg q \implies (p \vee r)$	$q \implies \neg r$
F	F	F	T	F	F
F	F	T	T	T	F
F	T	F	T	T	T
F	T	T	T	T	F
T	F	F	T	T	F
T	F	T	T	T	F
T	T	F	F	T	T
T	T	T	F	T	F

- Since at least one truth assignment simultaneously satisfies all equations, namely $p := F, q := T, r := F$, **this set of requirements is consistent.**
- In the language of this problem, that means the message is received and is not a draft and is not in the outbound queue



Solving Logic Puzzles

- Many puzzles are based around logical arguments and reasoning.
- For solution to such puzzles, propositional logic is a very useful tool.
- The idea is to model the elements of the puzzle as propositional formulae and use those formulas to determine if they are **satisfiable**, **consistent etc**, depending on the puzzle.

Example 1.5: Knights and Knaves

Suppose you are on an island with two kinds of people

- ① *Knights*, who always tell the truth; and
- ② *Knaves*, who always lie

You meet two people A and B. A says "B is a knight." B says "the two of us are different kind of people". **What kind of people are A and B**



Knights and Knaves

- **A and B are both Knaves**
- Let p "A is a knight", and q be "B is a knight".
- The statement "the two of us are different kind of people" can be represented as $(p \wedge \neg q) \vee (\neg p \wedge q)$
 - a Assume A is a knight and therefore p is true. Then, the statement "B is a knight" must be true and therefore q is true. If B is a knight, then the formula $(p \wedge \neg q) \vee (\neg p \wedge q)$ should be true. However, this formula is false if both p and q are true
 - b Assume A is a knave (i.e. $\neg p$ is true). Then, the statement "B is a knight" must be false (i.e. $\neg q$ is true). Therefore, B is a knave and the statement should be a lie. Both p and q being false makes $(p \wedge \neg q) \vee (\neg p \wedge q)$ false, which is **consistent**



Logical Equivalences

- Propositional logic is the foundation of more complex logical systems and formal mathematical proofs.
- Of particular importance is the *equivalence proof* that proves one thing is equivalent to another through a sequence of equivalences.

Definition 1.5: Tautology

A propositional formula that is always true, for every possible truth assignment

- A proposition that is not a tautology is either a contradiction or contingency

Definition 1.6: Contradiction

A proposition formula that is always false, for every possible truth assignment, is a contradiction

Definition 1.7: Contingency

A proposition formula that is neither a contradiction nor a tautology is a contingency



Logically Equivalent

- Two propositions formulas say p and q , are logically equivalent if $p \iff q$ is a tautology. If this is the case, we may write $p \iff q$ or $p \equiv q$
- An explicit way to determine if two expressions are logically equivalent is if their two columns in a truth table are identical
- Logically equivalent truth table

p	q	$\neg p$	$p \implies q$	$\neg p \wedge q$
F	F	T	T	T
F	T	T	T	T
T	F	F	F	F
T	T	F	T	T



Logically Equivalent

- Following are logical equivalences which should be memorized. They are similar to many laws of arithmetic

Law	$\wedge$	$\vee$
Identity	$p \wedge T \equiv p$	$p \vee F \equiv p$
Annihilation	$p \wedge F \equiv F$	$p \vee T \equiv T$
Idempotent	$p \wedge p \equiv p$	$p \vee p \equiv p$
Complementation	$p \wedge \neg p \equiv F$	$p \vee \neg p \equiv T$
Double Negation	$\neg(\neg p) \equiv p$	



Logically Equivalent

- Other interesting properties of logical connectives between two or more propositional variables

Law	$\wedge$ connective	$\vee$ connective
Commutativity	$p \wedge q \equiv q \wedge p$	$p \vee q \equiv q \vee p$
Associativity	$p \wedge (q \wedge r) \equiv (q \wedge p) \wedge r$	$p \vee (q \vee r) \equiv (q \vee p) \vee r$
Distributivity	$p \wedge (q \wedge r) \equiv (p \wedge q) \vee (p \wedge r)$	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
Absorption	$p \wedge (p \vee q) \equiv p$	$p \vee (p \wedge q) \equiv p$
De Morgan's Laws	$\neg(p \wedge q) \equiv \neg p \vee \neg q$	$\neg(p \vee q) \equiv \neg p \wedge \neg q$

- Demo Morgan's Laws are very important as they explain how negation distributes over conjunction and disjunction. It turns disjunction into conjunctions and vice versa



Logical Equivalences

- In these laws, we can replace any propositional variable with an entire propositional formula WLOG
- Let $p := (\neg a \wedge b) \vee c$

$$\begin{aligned}
 (\neg a \wedge b) \vee c \vee T &\equiv ((\neg a \wedge b) \vee c) \vee T \\
 &\equiv p \vee T \\
 &\equiv T
 \end{aligned}$$

- These laws and properties are used as building blocks to prove that certain expressions are logically equivalent. More logical equivalences

$$\begin{aligned}
 (p \implies q) &\equiv \neg p \vee q \\
 (p \implies q) \wedge (p \implies r) &\equiv p \implies (q \wedge r) \\
 (p \implies r) \wedge (q \implies r) &\equiv (p \vee q) \implies r \\
 p \iff q &\equiv (p \implies q) \wedge (q \implies p) \\
 p \iff q &\equiv \neg p \iff \neg q \\
 p \iff q &\equiv (p \wedge q) \vee (\neg p \wedge \neg q)
 \end{aligned}$$

- Draw truth table for the above propositions to prove that they are logically equivalent



Equivalence Proof

- To show two expressions say A and B , are logically equivalent we construct a sequence of logically equivalent statements E_i beginning with A and ending with B

$$A \equiv E_1$$

$$E_1 \equiv E_2$$

$$E_2 \equiv E_3$$

.

.

.

$$E_n \equiv B$$



Equivalence Proof - Example 1

- Prove that $p \wedge (\neg p \vee q) \equiv p \wedge q$

$$\begin{aligned} p \wedge (\neg p \vee q) &\equiv p \wedge \neg p \vee p \wedge q && \text{distributivity} \\ &\equiv F \vee p \wedge q && \text{complementation} \\ &\equiv p \wedge q && \text{Identity} \end{aligned}$$

- Prove that $p \vee (p \wedge q) \vee (\neg p \wedge r) \equiv (p \wedge q) \vee p \vee r$

$$\begin{aligned} p \vee (p \wedge q) \vee (\neg p \wedge r) &\equiv (p \wedge q) \vee p \vee (\neg p \wedge r) && \text{commutativity} \\ &\equiv (p \wedge q) \vee (p \vee \neg p) \wedge (p \vee r) && \text{distributivity} \\ &\equiv (p \wedge q) \vee (T \wedge (p \vee r)) && \text{Complementation} \\ &\equiv (p \wedge q) \vee (p \vee r) && \text{Identity} \end{aligned}$$



Exercises

- Classify each of the following statements as True or False
 - ① $4 = 2 + 2$ and $7 < \sqrt{50}$
 - ② $4 = 2 + 2 \rightarrow 7 < \sqrt{50}$
 - ③ $4 \neq 2 + 2 \rightarrow 7 < \sqrt{50}$
 - ④ $4 = 2 + 2 \rightarrow 7 \geq \sqrt{50}$
- Give the negation of each of the following compound statements.
 - ① $a^2 > 0$ or a is not a real number
 - ② $x = \pm 1$
 - ③ x is a real number and $x^2 + 1 = 0$
- Give the converse, inverse, and contrapositive of each of the following implications
 - ① If $\frac{a}{b}$ and $\frac{b}{c}$ are integers, then $\frac{a}{c}$ is an integer
 - ② $ab = 0 \rightarrow a = 0$



Exercises

- For each of the following propositional formulas, determine which are satisfiable. If they are satisfiable, give a truth assignment which satisfies the formula.

① $p \wedge (\neg q \vee \neg r) \wedge (q \vee \neg p)$

② $p \wedge (q \vee \neg p) \wedge (\neg q \vee \neg p) \wedge r$

③ $(p \wedge q \wedge \neg r) \vee (p \wedge \neg q \wedge \neg r)$



Application Exercise

Define the basic clauses of each of the following statements as propositional variables. Then, express each of the following compound statements as propositional formulas. Finally, is this set of propositions consistent? If so, give a truth assignment which shows they are consistent.

- ① The campus server does not work if the internet is off.
- ② Students can skype during the test when the Prof is distracted.
- ③ If the classroom phone does not ring then the Prof is not distracted.
- ④ Students cannot skype during the test unless the internet is on.
- ⑤ If the classroom phone rings then the campus server works.



Predicate Logic

- Propositional Logic is insufficient to represent more complex logical expressions. For example, the following cannot be represented.
- $p := \text{"All men are mortal"}$ and $q := \text{"Socrates is a man"}$. Is socrates mortal? The answer is yes.



Predicates and Variables

- Predicate logic augments propositional logic with variables, predicates, and quantifiers.
- Variables are simple. They are some symbol, for example x, y , or z that stands in for some mathematical object.

Definition 2.1: Predicate

A Symbol which represents a property or relation.

- Predicates are typically capital letters $P, Q, R, S \dots$
- Predicates stand for something. They are more general in that they need not have a truth value. Whereas propositions need to be declarative statement(i.e. an independent clause in a sentence), predicates can be an incomplete clause



Predicates and Variables

The following are all predicates

- D := "is a Dog"
- R := "is rowing his boat"
- P := "is a positive number"
- E := "is an even integer"

As can be observed in the above examples, predicates lack subjects. When Combined with variables, predicates allow us to form propositional functions

Definition 2.2: Propositional Functions

A function formed by a predicate and one or more variables.

A proposition is obtained by applying particular values to the variable(s) or by binding the variables with logical quantifiers.

- Propositional functions become propositions when their variables are replaced by a value in the domain of discourse or domain.



Propositional Function

- The domain specifies what are the possible values for the variable. a Domain is represented by U
- If there is no domain, propositional function has little meaning. Suppose $P(x)$ stands for "x passed ICS 2204", this only makes sense if the domain x is the "people who have taken ICS 2204"
- Let P be "is greater than 0". Then $P(x)$ is a propositional function denoting x is greater than 0. If we let the domain be the integers, x may take the value of any integer
 - ① $P(-3)$ is false
 - ② $P(0)$ is false
 - ③ $P(2)$ is true
 - ④ $P(3.14159)$ is invalid
- Replace a particular value from the domain of discourse for a variable in a propositional function, we say that the variable is bound - also called bound variable
- A variable that is not bound is called a free variable



Propositional Function - Example

- Let $R(x, y, z) := x + y = z$. Let $Q(x, y, z) := x - y = z$. Let U be the integers x, y , and z . What are the values of the following?
 - 1 $R(x, 3, z)$
 - 2 $R(2, -1, 5)$
 - 3 $R(3, 4, 7)$
 - 4 $Q(2, -1, 3)$
 - 5 $Q(3, 4, 7)$
 - 6 $Q(x, 3, z)$
- For compound expressions, logical connectives of propositional logic also apply to predicate logic. Examples
- Let $P(x) := x$ can fly. Let $Q(x) := x$ are blue. Let the domain of x be all species of animals
 - 1 $P(\text{penguins}) \vee P(\text{geese})$ is true
 - 2 $P(\text{blue jays}) \wedge Q(\text{blue jays})$ is true
 - 3 $P(\text{penguins}) \rightarrow Q(\text{tigers})$



Propositional Function - Example

- Let P, Q be predicates. Let x, y be variables
 - 1 $P(2) \vee Q(5)$ is a proposition since all variables are bound
 - 2 $P(3) \wedge P(x)$ is a propositional function with one variable
 - 3 $P(x) \rightarrow P(y)$ is a propositional function with two variables
 - 4 $P(y) \vee Q(x)$ is a propositional function with two variables



Quantifiers

- Quantifiers are used as modifiers to variables - gives rise to the logical formalization of the English words "all", "some", "any", "every", etc.

Example 2.1

- ① all men are mortal
- ② some birds cannot fly

- These cannot be expressed in propositional logic but can be represented in predicate logic using quantifiers.
- The two main quantifiers are the **universal quantifier** and the **existential quantifier**



Quantifiers

Definition 2.3: Universal Quantifier

"For all", "For Any", "Given any". A Universal quantifier asserts that a predicate is true for any and every value in the domain. Denoted by $\forall$

Definition 2.4: Existential Quantifier

"There exists", "For some", "There is a", "There is at least one". An existential quantifier asserts that a predicate is true for at least one value in the domain - Denoted $\exists$

A quantifier binds a variable to a value just like substituting a variable to a value from a domain.



Universal Quantifier

- Let P be a predicate and U be the domain.
 - $\forall x P(x)$ says that $P(x)$ is true for *every choice* of x from the domain
 - $\exists x P(x)$ says that $P(x)$ is true for *some* x in the domain
- $\forall x P(x)$ can be read as "For all x , $P(x)$ "

Example 2.2

- Let $P(x) := x > 0$. If U is all integers, $\forall x P(x)$ is *false*
- Let $P(x) := 2x$ is even. If U is all integers, $\forall x P(x)$ is *true*
- Let $P(x) := x$ has feathers. If U is all animals, $\forall x P(x)$ is *false*
- Let $P(x) := x > 0$. If U is all integers, $\forall x P(x)$ is *false*



Existential Quantifier

$\exists x P(x)$ can be read as "There exists an x such that $P(x)$ "

Example 2.3

- a Let $P(x) := x > 0$. If U is all integers, $\exists x P(x)$ is true
- b Let $P(x) := 2x$ is odd. If U is all integers, $\exists x P(x)$ is false
- c Let $P(x) := x$ has feathers. If U is all animals, $\exists x P(x)$ is true
- d Let $P(x) := x$ has feathers. If U is all birds, $\exists x P(x)$ is true

Note that the truth value of a propositional function whose variable is bound by a quantifier depends on the particular domain of discourse



Exercise

Let $P(x) := x < 2$

Example 2.4

- i If U is positive integers, what is the truth value of $\forall x P(x)$?
What is the value of $\exists x P(x)$
- ii If U is negative integers, what the truth value of $\forall x P(x)$?
What is the truth value of $\exists x P(x)$?
- iii If U is the numbers 3, 4, 5 what is the truth value of $\forall x P(x)$?
What is the truth value of $\exists x P(x)$



Let $P(x)$ be a propositional function with a universe U

- ① If $\forall x P(x)$ is true, then $\exists x P(x)$ *must* be true.
- ② If $\exists x P(x)$ is false, then $\forall x P(x)$ must be false.

That is $\forall x P(x) \rightarrow \exists x P(x)$ and $\neg(\exists x P(x)) \rightarrow \neg(\forall x P(x))$



Precedence of Quantifiers

Quantifiers have a higher precedence than all other logical connectives

- $\exists x P(x) \vee Q(x) \equiv (\exists x P(x)) \vee Q(x)$
- $\exists x P(x) \wedge Q(x) \equiv (\exists x P(x)) \wedge Q(x)$
- $\forall x P(x) \rightarrow Q(x) \equiv (\forall x P(x)) \rightarrow Q(x)$
- $P(x) \rightarrow \exists x Q(x) \equiv P(x) \rightarrow (\exists x Q(x))$



Quantified Compounds

Let $P(x) := "x \text{ is a son}"$

Let $Q(x) := "x \text{ is a parent}"$

Let U be all people

- $\forall x P(x) \rightarrow Q(x)$ translates to "if every person has a son, then x is a parent" i.e we have *not bound the variable* in the predicate $Q(x)$
- $\forall x (P(x) \rightarrow Q(x))$ translates to "For all people, if a person has a son then that person is a parent"



Negating Quantifiers

This must be done carefully to avoid losing the meaning of a sentence

Example 2.5: Negating English

- ① "Every student in this class knows Python"
 - Ⓐ "It is not the case that every student in this class knows Python" or
 - Ⓑ "Not every student in this class knows Python" or
 - Ⓒ "At least one student in this class does not know Python"
- ② "Someone thinks football is the best sport"
 - ① "It is not the case that someone thinks football is the best sport" or
 - ② "No one this football is the best sport" or
 - ③ "Everyone thinks football is not the best sport"



As above, quantifiers can be negated in two ways

- Negate the quantifier itself
- negate the Inner predicate

This is similar to Demorgan's law for propositional logic. Negating a universal or existential quantifier "flips" it become the other one

Example 2.6: Illustration

$$\neg(\forall x P(x)) \equiv \exists x \neg P(x)$$

$$\neg(\exists x P(x)) \equiv \forall x \neg P(x)$$

Example 2.7

Let $P :=$ "Knows Python" with U being all students in this class

$$\neg(\forall x P(x)) \equiv \exists x \neg P(x)$$

There exists a student in this class who does not know Python

Let $S :=$ "Thinks football is the best sport" with U being all people

$$\neg(\exists x S(x)) \equiv \forall x \neg S(x)$$

Every one does not this football is the best sport



Truth and Fallacy of Quantifiers

Quantified Expression	Equivalent Expression	When True?	When False?
$\forall x P(x)$		When $P(x)$ is true for every possible x	When $P(x)$ is false for at least one possible x
$\exists x P(x)$		When $P(x)$ is true for at least one possible x	When $P(x)$ is false for every possible x
$\neg \forall x P(x)$	$\exists x \neg P(x)$	When $P(x)$ is false for at least one possible x	When $P(x)$ is true for every possible x
$\neg \exists x P(x)$	$\forall x \neg P(x)$	When $P(x)$ is false for every possible x	When $P(x)$ is true for at least one possible x

Try to internalize the subtle differences in each case. They are important throughout the course and in the future.



Exercises

Translate the following sentences to predicate logic, using quantifiers as appropriate. Can you discern the truth value from the resulting expressions?

- ① "Some student in this class has visited Mecca"
- ② "Every Student in this class has visited Us or Mecca"
- ③ "An Animal that can fly is an animal with wings"



Exercise - Solution

Translate the following sentences to predicate logic, using quantifiers as appropriate. Can you discern a truth value from the resulting expression?



Exercise - Solution

Translate the following sentences to predicate logic, using quantifiers as appropriate. Can you discern a truth value from the resulting expression?

Let M : = "Has visited Mecca";



Exercise - Solution

Translate the following sentences to predicate logic, using quantifiers as appropriate. Can you discern a truth value from the resulting expression?

Let M : = "Has visited Mecca"; Let S : = "Has visited US";



Exercise - Solution

Translate the following sentences to predicate logic, using quantifiers as appropriate. Can you discern a truth value from the resulting expression?

Let M : = "Has visited Mecca"; Let S : = "Has visited US"; Let W : = "Has wings";



Exercise - Solution

Translate the following sentences to predicate logic, using quantifiers as appropriate. Can you discern a truth value from the resulting expression?

Let M : = "Has visited Mecca"; Let S : = "Has visited US"; Let W : = "Has wings"; Let Y : = "Can fly";



Exercise - Solution

Translate the following sentences to predicate logic, using quantifiers as appropriate. Can you discern a truth value from the resulting expression?

Let M : = "Has visited Mecca"; Let S : = "Has visited US"; Let W : = "Has wings"; Let Y : = "Can fly";

① "Some student in this class has visited Mecca"

- If U is students in this class, then: $\exists x M(x)$
- If U is all people, let T : = "is a student in this class". Then we have $\exists x T(x) \rightarrow M(x)$
- I personally do not know the truth value of this. However, if you are reading this and you have been to Mecca, then your existence in this class provides an "existence proof" that the statement is true.



Exercise - Solution

② "Every Student in this class has visited Us or Mecca"

- If U is student in this class: $\forall x (M(x) \vee S(x))$
- If U is all people: $\forall x [T(x) \rightarrow (M(x) \vee S(x))]$
- The truth value of this statement is true. You are all attending school in US, and have been to campus in London, Ontario. The statement is true without even knowing the truth value of $M(x)$



Exercise - Solution

③ "An Animal that can fly is an animal with wings"

- Let U be all animals $\forall x (W(x) \rightarrow Y(x))$
- Notice that this English statement does not explicitly use "every" or "for all". However, it is still a statement which is stating a universality. It is stating that any animal that can fly must have wings. Therefore we use the universal quantifier.
- The truth value of this statement is true. All animals which fly are animals which have wings. Notice that the converse is not true (penguins have wings but cannot fly). Can you prove me wrong? Do you know of an animal that can fly but doesn't have wings? If you can identify such an animal, that would be a "proof by counterexample".



Nested Quantifiers

Nesting quantifiers are necessary for natural language with several clauses and variables. It has natural extension to math and computer science

Example 2.8: Nested Quantifier

"Every real number has a multiplicative inverse"

Let U be the set of real numbers

$$\forall x \exists y (x \times y = 1)$$

Nested Quantifiers can be thought of as Propositional functions.

From the above example, let $P(x, y) := (x \times y = 1)$ and $Q(x) := \exists y P(x, y)$ can be interpreted that x and y are multiplicative inverses, meanwhile $Q(x)$ mean x has a multiplicative inverse.



Nested Quantifiers

Then, "every real number has a multiplicative inverse" can be written as:

$$\begin{aligned}\forall x Q(x) &\equiv \forall x (\exists y P(x, y)) \\ &\equiv \forall x (\exists y (x \times y = 1)) \\ &\equiv \forall x \exists y (x \times y = 1)\end{aligned}$$

- The best way to think about nested quantifiers is that variables are bound or "fixed" left to right. In $\exists x \forall y P(x, y)$, x is bound first, regardless of what happens to y . Then y is bound
- Another way to think about nested quantifiers is nested loops.



Consider the following loop and subsequent expressions

```
for i in range(n) :  
    for j in range(m) :  
        P(i , j)
```

Expression	When is it true?
$\forall i \forall j P(i,j)$	Only true if all iterations of both loops are true.
$\forall i \exists j P(i,j)$	True when, for every iteration of the outer loop, at least one iteration of the inner loop is true.
$\exists i \forall j P(i,j)$	Is true when, for at least one iteration of the outer loop, every iteration of the inner loop is true.
$\exists i \exists j P(i,j)$	Is true if at least one iteration of the inner loop is true, for any iteration of the outer loop.

Quantifiers to English

Example 2.9

Let $C(x) := x$ has a computer,



Quantifiers to English

Example 2.10

Let $C(x) := x$ has a computer, $F(x, y) := x$ and y are Friends, where U is all students at Western

$$\textcircled{1} \quad \forall x (C(x) \vee \exists y (C(y) \wedge F(x, y)))$$

In English, this means "Every student at Western has a computer or Has a friend with a computer"

$$\textcircled{2} \quad \exists x \forall y \forall z ((F(x, y) \wedge F(x, z) \wedge (y \neq z)) \rightarrow \neg F(y, z))$$

"There exists a student with two different friends who are not friends with each other."



A large part of being able to understand, and translate, mathematical statements comes with practice. But, there is a general methodology we can use to convert mathematical statements to logic.

- 1 Write the statement to be more verbose. Replace phrases like " q whenever p " with "if p then q " to bring the statement closer to direct translation.
- 2 Explicitly introduce variables. Replace phrases like "a number greater than two" with "a number x such that $x > 2$ "
- 3 Finally, translate the sentence to a logical expression using quantifiers, predicates, and logical connectives.



Example 2.11

Write "the sum of two positive integers is always positive" as a logical expression.

- 1 Make the sentence be more verbose: "for any two integers, if they are both positive, then their sum is positive"
- 2 Use variables: "for any two integers x and y , if x and y are both positive, then their sum is positive"
- 3 Translate to a logical expression

$$\forall x \forall y ((x > 0) \wedge (y > 0) \rightarrow (x + y > 0)),$$

where U is all integers

Revisiting System Specifications

Predicate Logic is exceptionally useful for system specifications. This application requires special care with nested quantifiers and precedence between quantifiers and logical connectives.

Example 2.12: System Specification with Predicates

Consider a mail-server with limited bandwidth. The designer specifies three requirements

- ① "Every message larger than 1 megabyte must be compressed."
- ② "Messages larger than 10 megabytes cannot be sent."

The domain is left implicit since it is obvious Let $L(m, y) :=$ "message m is larger than y megabytes"

Revisiting System Specifications

Predicate Logic is exceptionally useful for system specifications. This application requires special care with nested quantifiers and precedence between quantifiers and logical connectives.

Example 2.13: System Specification with Predicates

Consider a mail-server with limited bandwidth. The designer specifies three requirements

- ① "Every message larger than 1 megabyte must be compressed."
- ② "Messages larger than 10 megabytes cannot be sent."

The domain is left implicit since it is obvious Let $L(m, y) :=$ "message m is larger than y megabytes" Let $C(m) :=$ "message m is compressed"



Revisiting System Specifications

Predicate Logic is exceptionally useful for system specifications. This application requires special care with nested quantifiers and precedence between quantifiers and logical connectives.

Example 2.14: System Specification with Predicates

Consider a mail-server with limited bandwidth. The designer specifies three requirements

- ① "Every message larger than 1 megabyte must be compressed."
- ② "Messages larger than 10 megabytes cannot be sent."

The domain is left implicit since it is obvious Let $L(m, y) :=$ "message m is larger than y megabytes" Let $C(m) :=$ "message m is compressed" Let $S(u, m) :=$ "user u can send message m "

- ① $\forall m (L(m, 1) \rightarrow C(m))$
- ② $\forall m (L(m, 10) \rightarrow \forall u \neg S(u, m)) \equiv \forall m (L(m, 10) \rightarrow \neg \exists u S(u, m))$



Possible Pitfalls

Predicates, quantifiers, and logical connectives have a strict order of precedence that can drastically change their interpretation. When in doubt, use parentheses!

We know from Negating Quantifiers that negation “flips” a universal quantifier or existential quantifier to become the other. But what happens when we have nested quantifiers? Simply use parentheses and proceed one at a time.

$$\begin{aligned}\neg \forall x \exists y \forall z P(x, y, z) &\equiv \neg (\forall x (\exists y (\forall z P(x, y, z)))) \\ &\equiv (\exists x \neg (\exists y (\forall z P(x, y, z)))) \\ &\equiv (\exists x (\forall y \neg (\forall z P(x, y, z)))) \\ &\equiv (\exists x (\forall y (\exists z \neg P(x, y, z))))\end{aligned}$$



Quantifiers and Connectives

Quantifiers have a higher precedence than logical connectives.

Parenthesis are used to apply a quantifier to a compound expression.

Is it possible to “distribute” a quantifier into a compound expression as we did with negation? The answer is sometimes. The following is valid

$$\forall x (P(x) \wedge Q(x)) \equiv \forall x P(x) \wedge \forall x Q(x)$$

$$\exists x (P(x) \vee Q(x)) \equiv \exists x P(x) \vee \exists x Q(x)$$

Example 2.15

To see why the above is: let the domain of x be a, b, c

$$\begin{aligned}\forall x (P(x) \wedge Q(x)) &\equiv (P(a) \wedge Q(a)) \wedge (P(b) \wedge Q(b)) \wedge (P(c) \wedge Q(c)) \\ &\equiv P(a) \wedge P(b) \wedge P(c) \wedge Q(a) \wedge Q(b) \wedge Q(c) \\ &\equiv \forall x P(x) \wedge \forall x Q(x)\end{aligned}$$



Quantifiers and Connectives

Example 2.16

To see why the above is: let the domain of x be a, b, c

$$\begin{aligned}\exists x (P(x) \vee Q(x)) &\equiv P(a) \vee P(b) \vee P(c) \vee Q(a) \vee Q(b) \vee Q(c) \\ &\equiv P(a) \vee P(b) \vee P(c) \vee Q(a) \vee Q(b) \vee Q(c) \\ &\equiv \exists x P(x) \vee \exists x Q(x)\end{aligned}$$

However, in more cases, quantifiers do not distribute. Here are some counterexamples.

Example 2.17

$$\exists x (P(x) \wedge Q(x)) \not\equiv \exists x P(x) \wedge \exists x Q(x)$$

$$\forall x (P(x) \vee Q(x)) \not\equiv \forall x P(x) \vee \forall x Q(x)$$

$$\exists x (P(x) \rightarrow Q(x)) \not\equiv \exists x P(x) \rightarrow \exists x Q(x)$$

$$\forall x (P(x) \rightarrow Q(x)) \not\equiv \forall x P(x) \rightarrow \forall x Q(x)$$



Quantifiers and Connectives

The major reason why existential quantifiers do not distribute is because the two quantifiers bind the variables in different ways.

$\exists x P(x) \wedge \exists x Q(x)$ means "There exists some x such that $P(x)$ AND there exists some x such that $Q(x)$ "

The x in the first clause *is not* the same x in the second clause. x was quantified twice in different contexts.

The above statement is very different from $\exists x(P(x) \wedge Q(x))$ which means "There exists some x such that $P(x)$ and $Q(x)$ are true".

In this case, the same object is being applied simultaneously to P and Q



Predicates and Variables

- ① Use the appropriate quantifiers "for all" and "there exists" to write the following sentences
 - a "Not all continuous functions are differentiable."
 - b There is no largest real number."
 - c "Every positive integer is the product of some collection of primes."
- ② Use the appropriate quantifiers "for all" and "there exists" to write the following sentences. You don't have to translate to predicate logic, just re-write in English being more verbose and including the phrases "for all" and "there exists" as appropriate.
 - a A polynomial with integer coefficients may have integer roots.
 - b A polygon may not be both convex and concave.
 - c Trigonometric functions are periodic.



Quantifiers

Let $F(x) := x$ is fish

Let $H(x) := x$ is whale

Let $W(x) := x$ inhabits the water

Let $S(x) := x$ has scales

If the domain of x is all animals, what is the truth value of the following?

- i $\forall x (F(x) \rightarrow W(x))$
- ii $\exists x (S(x) \rightarrow \neg W(x))$
- iii $\exists x (F(x) \wedge \neg S(x) \wedge W(x))$
- iv $\forall x S(x) \rightarrow \forall x W(x)$



Let $B(x) :=$ " x is a bird"

Let $E(x) :=$ " x lays eggs"

Let $M(x) :=$ " x is a mammal"

Let $P(x) :=$ " x is a plant"

Let $W(x) :=$ " x inhabits the water"

If the domain of x is all plant and animals, translate each of these quantified statements to a natural language statement. What is the truth value of each of them

- ① $\forall x (B(x) \rightarrow E(x))$
- ② $\forall x (E(x) \rightarrow B(x))$
- ③ $\exists x (W(x) \wedge P(x))$
- ④ $\exists x (W(x) \wedge E(x))$
- ⑤ $\forall x (P(x) \vee M(x) \vee B(x))$
- ⑥ $\forall x (W(x) \rightarrow \neg M(x))$



Quantifier Solutions

- ① **True.** All birds lay eggs.
- ② **False.** Not all egg-layers are birds. Reptiles, for example.
- ③ **True.** There are water plants. For example, seaweed.
- ④ **True.** Egg-laying water dwellers? Puffins, Penguins, for example.
- ⑤ **False.** There are organisms which are neither plants, mammals or birds. Reptiles, amphibians, etc.
- ⑥ **False.** There are water-inhabiting organisms which are mammals. Whales!



Quantifiers - Exercises Cont'd

Let U be the set of imaginary creatures consisting drackles, klabs and grebbits

$D(x) := x$ is a drackle

$K(x) := x$ is a klab

$G(x) := x$ is a grebbit

Translate the following into predicate logic statements

- ① Everything is a drackle.
- ② Nothing is a klab.
- ③ All drackles are klabs.
- ④ Some drackles are grebbits.
- ⑤ No klab is a grebbit.



Quantifier Solutions

- ① $\forall x D(x)$
- ② $\neg \exists x K(x) \equiv \forall x \neg K(x)$
- ③ $\forall x (D(x) \rightarrow K(x))$
- ④ $\exists x (D(x) \wedge G(x))$
- ⑤ $\neg \exists x (K(x) \wedge G(x)) \equiv \forall x (\neg K(x) \vee \neg G(x))$



Application Exercises

Consider the following requirements:

- a “If a user is active, then they are connected to the server”
- b “Only users connected to the server can receive messages”
- c “A user that requests encryption must receive encrypted messages”

Let the domain of u be all users of the system and the domain of e be “encrypted” or “unencrypted”.

Let

Predicate	Description
$A(u) := u$ is an active user	$C(u) := u$ is an connected to the server
$R(u, e) := u$ can receive messages of type e	$E(u) := u$ has requested encryption

Translate these requirements to quantified predicates. Is that set of propositions consistent? If so, give a satisfying truth assignment. If not, what can you add to one of these requirements (using the already supplied predicates) to make the requirements consistent?



Application Solution

- a $\forall u (A(u) \rightarrow C(u))$
- b $\forall u \exists e (C(u) \leftrightarrow R(u, e))$
- c $\forall u (E(u) \rightarrow R(u, \text{encrypted}) \wedge \neg R(u, \text{unencrypted}))$

These are not consistent. If $A(u)$, $C(u)$ are false and $E(u)$ is true, then the third requirement implies that u can receive messages, contradicting (b)

Modify the third requirement:

$$\exists u (E(u) \wedge C(u) \rightarrow R(u, \text{encrypted}) \wedge \neg R(u, \text{unencrypted}))$$



Quantifier Exercises

Let $T(x, y)$ be the predicate "x is taking y" where the domain of x is all students and the domain of y is all courses

Translate the following English statements into quantified predicate statements

- ① "Every course is being taken by at least one student"
- ② "Some student is taking every course"
- ③ "No student is taking all courses"
- ④ "There is a course that all students are taking"
- ⑤ "Every student is taking at least one course"
- ⑥ "There is a course that no students are taking"
- ⑦ "Some students are taking no courses"
- ⑧ "No course is being taken by all students"
- ⑨ "Some courses are being taken by no students"
- ⑩ "No student is taking any course"



Quantifier Solutions

- 1 $\forall C \exists S T(S, C)$
- 2 $\exists S \forall C T(S, C)$
- 3 $\forall S \exists C \neg T(S, C)$
- 4 $\exists C \forall S T(S, C)$
- 5 $\forall S \exists C T(S, C)$
- 6 $\exists C \forall S \neg T(S, C)$
- 7 $\exists S \forall C \neg T(S, C)$
- 8 $\forall C \exists S \neg T(S, C)$
- 9 $\exists C \forall S \neg T(S, C)$
- 10 $\forall S \forall C \neg T(S, C)$



Having solid understanding of propositional and predicate logic, the section seeks to put them into practice. We will examine a methods of proofs and strategies to formally prove something.

Recall equivalence proofs as one kind of proof. "Proof by truth table" is a valid option for proving equivalence.

This section examines prove of general statements that cannot be easily be proved by a truth table.



Definition 3.1: Proof

A proof is a precise, formal, and valid argument which establishes the truth of a statement

Just like how **Satisfiability and Consistency** have a direct applications to computer science, so too do mathematical proofs

Example 3.1: Uses of Proof

- 1 Computer and Software System Verification
- 2 Cybersecurity and Cryptography
- 3 Artificial Intelligence; and
- 4 Guaranteeing algorithm Correctness



Provable Things

Theorem 3.1

A **theorem** is a formal statement that be shown to be true. It establishes a truth from which other theorems can be proved, or algorithm constructed

Lemma 3.1

A **lemma** is a formal statement that can be shown to be true, and is used to proof another theorem. It is a helper "theorem"

Proposition 3.1

A **proposition** is a formal statement that can be shown to be true, but the importance of this fact is less than that of a theorem



Provable Things

Corollary 3.1

A **corollary** is a special case of a theorem, lemma, or proposition. It follows directly from a theorem and is not typically proved explicitly. Corollaries are stated to show an “important result” or “important implication” of a theorem.

Conjecture 3.1: Conjecture

A **Conjecture** is a formal statement that is likely to be true, but has not yet been proved to be true. Once proven to be true, it may become a formal theorem or lemma.

Whether its a theorem, a lemma or a proposition, the way they are stated and the way they are proved remains the same. For consistency's sake, a theorem will be used.



Theorem Statements

Theorems are typically stated as inferences or byconditionals. "If P then Q " or " P if and only if Q ". The structure of using "let x be $\dots$ " is an implicit *universal quantifier*.

For all x which satisfy this property, the inference (theorem) holds.

With a little practice, we can extract the implication from such statements.



Theorem Statements

Example 3.2: Implicit Implications

Consider the following theorem statements

Theorem 3.2

Let x, y be positive real numbers such that $x > y$. Then, $x^2 > y^2$

This also means

Theorem 3.3

If $x > y$, where x and y are positive numbers, then $x^2 > y^2$

Which also mean

Theorem 3.4

"For all positive real numbers x and y , if $x > y$ then $x^2 > y^2$ "



In the example 3.2, the closest to being a direct translation is the last statement.

Let $P(x, y) := x > y$

Let $Q(x, y) := x^2 > y^2$

If we let the domain of x and y be all positive real numbers, then the previous example states

$$\forall x \forall y (P(x, y) \rightarrow Q(x, y))$$



Trivial Proof

Proposition 3.2

If it is raining then $1 = 1$

Proof.

This implication is trivially true since $1 = 1$ is known to be true. The truth value of the premise does not matter □

Proof for implications $p \rightarrow q$ relies on p being false. Recall "Falsity implies" anything. If we know p to be false, then the implication $p \rightarrow q$ is always true. A **vacuous proof** is thus a simple statement that shows p to be obviously false



Direct Proofs

- One of the most straight forward types of proofs.
- If we wish to prove the statement $p \rightarrow q$, we assume p is true and use laws and axioms and logical equivalence to show that q must also be true
- Also direct proof can be defined as follows: if we prove a statement $A \rightarrow B$, we assume A is true and then prove a sequence of implications:
 $A \rightarrow A_1 \rightarrow A_2 \rightarrow \dots \rightarrow B$ for some statements $A_1, A_2, \dots$

Definition 3.2: Even and Odd Integers

An integer m is even if it satisfies the formula $m = 2k$ for some integer k . An integer n is odd if it satisfies the formula $n = 2k + 1$ for some integer k

**Definition 3.3: Rational Number**

A number n is *rational* if there exists integer p and q such that $n = \frac{p}{q}$ and $q \neq 0$

Using definitions 3.2 and 3.3 we used direct proof to prove the following proposition

Proposition 3.3

If n is an odd integer, then n^2 is odd

Proof.

Let $P(n) := "n \text{ is odd}"$, Let $Q(n) := "n^2 \text{ is odd}"$. Let the domain of n be all integers.

We look to prove $\forall n (P(n) \rightarrow Q(n))$

Assume $P(n)$ is true. $\therefore$ there exists some integer k such that $n = 2k + 1$.

Then, $n^2 = (2k + 1)^2 = (4k^2 + 4k + 1) = 2(2k^2 + 2k) + 1$.

Letting $k' = 2k^2 + 2k$, we have that k' is an integer and $n^2 = 2k' + 1$.

$\therefore n^2$ is odd





Example 3.3

Proposition 3.4

The sum of two rational numbers is rational

Proof.

Let $P(x) := "x \text{ is a rational number, where the domain of } x \text{ is all real numbers.}$

Proove that is needed: $\forall x \forall y (P(x) \wedge P(y) \rightarrow P(x + y))$

Assume that x and y are rational numbers.

Then, there exists integers a, b, c, d such that $x = \frac{a}{b}, y = \frac{c}{d}, b \neq 0$, and $d \neq 0$.

We have,

$$x + y = \frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd} = \frac{f}{g},$$

Where $f = ad + bc$ and $g = bd \neq 0$ are integers.

$\therefore x + y$ is rational





Proving Biconditionals

A theorem can also be stated as a biconditional. i.e. $p \leftrightarrow q$

In this case the prove contains two parts. Recall

$$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

Hence, we must prove two implications.

In this kind of proof, proving $p \rightarrow q$ is called "*proving the sufficient condition*"; and proving $q \rightarrow p$ is called "*proving the necessary condition*"



Proving Biconditionals

Prove the following statement using a direct proof

Example 3.4: Biconditional Prove - direct

Proposition 3.5

"A non-zero number q is rational if and only if there exists a non-zero integer n such that qn is an integer."



Solution to the proposition 3.5

Proof.

In this case the proof is biconditional, an "if and only if"

- ① "If q is a non-zero rational number then there exists a non-zero integer n such that qn is an integer", and
- ② "If qn is an integer for some number q and some non-zero integer n , then q is rational"

Proof of 1:

Assume q is rational. $\therefore, q = \frac{a}{b}$ for some integers a, b with $b \neq 0$. Take $n = b$. Then $nq = n\frac{a}{b} = a$. Since a is an integer, so is qn

Proof of 2:

Assume n is a non-zero integer and qn is an integer.

This means, there exists an integer r such that $qn = r$

Since n is non-zero, we can divide both sides by n resulting to $q = \frac{r}{n}$.

As both r and n are integers, q is a rational number □



For this kind proofs, we prove a statement like $A \rightarrow B$ without following a sequence of implications $A \rightarrow A_1 \rightarrow \dots \rightarrow B$. Instead we proceed in an unexpected way.

There are two main types of indirect proofs

- ① Proof by Contrapositive
- ② Proof by Contradiction



Proof by Contrapositive

Proving a statement $p \rightarrow q$ requires that that we prove its contrapositive.

Thus the statement $\neg q \rightarrow \neg p$ which is its contrapositive equivalent.

Recall an **implication** and its **contrapositive** are **equivalent**, that is $p \rightarrow q \equiv \neg q \rightarrow \neg p$.

Proving the Contrapositive can be done using any other proof method.

Typicall this is done by direct method.

However, this is not always true



Proof by Contrapositive

Example 3.5: Proof by Contrapositive

Proposition 3.6

If n is an integer such that $3n + 2$ is odd, then n is odd

Proof.

Proceed the proof by contrapositive.

Assume n is not odd.

Then we prove that $3n + 2$ is not odd.

Since n is not odd, then it is even, therefore $n = 2k$ for some integer k

Substituting:

$$3n + 2 = 3(2k) + 2 = 6k + 2 = 2(3k + 1)$$

Let $k' = 3k + 1$, then $3n + 2 = 2k'$

Hence, it is even. □



Proof by Contradiction

For this method of proof, we look to assume some fact and then find a contradiction.

Recall that a contradiction is a statement that is **always false**

Suppose we want to prove the statement p .

By contradiction, we assume $\neg p$ is true, and derive a contradiction.

That is, we show that $\neg p \rightarrow A_1 \rightarrow A_2 \rightarrow \dots \rightarrow \mathbf{F}$ for some statement $A_1, A_2, \dots$

By contrapositive, if $\neg p \rightarrow \mathbf{F}$ holds then $\mathbf{T} \rightarrow p$ holds.

$\therefore p$ must be true.



Example 3.6: Proof by Contradiction

Theorem 3.5

There is no largest integer

Proof.

Proceed the prove by contradiction

Assume that there does exist some largest integer called n .

But $n + 1$ is also an integer and $n + 1$ is strictly larger than n .

$\therefore$ our assumption that a largest integer exist is false.

That is, there is no largest integer is true. □



Strategies to Help with Proofs

This section guides in helping writing and deciphering proofs. These strategies can be incorporated into various different proof methods

- 1 Finding errors in proofs
- 2 Proof by cases
- 3 Without loss of generality
- 4 Disproof by counterexample



Finding errors in proofs

Recall, a proof is a logical and sequential argument.

A Proof only holds if every step along the way is correct.

A single "misstep" makes the proof invalid and thus not proving anything.

Find errors in the following proposition

Proposition 3.7

$$1 = 2$$

Proof.

1.	$a = b$	Premise
2.	$a^2 = ab$	Multiple both sides by a
3.	$a^2 - b^2 = ab - b^2$	Subtract b^2 from both sides
4.	$(a - b)(a + b) = b(a - b)$	Factor both sides
5.	$a + b = b$	Divide both sides by $a - b$
6.	$2b = b$	Since $a = b$, $a + b = 2b$
7.	$2 = 1$	Divide both sides by b





Proof by Cases

Where a theorem premise is composed of a compound statement, it is recommended that you prove all the possible cases. e.g. Let n be an integer. In this case, n could be negative, zero or positive.

There are two steps in a proof by cases:

- 1 Extract from the theorem statement the possible cases; and
- 2 Prove each case independently

The validity of proof by cases come from the following logical equivalence

$$(p_1 \vee p_2 \vee \cdots \vee p_n) \rightarrow q \equiv (p_1 \rightarrow q) \wedge (p_2 \rightarrow q) \wedge \cdots \wedge (p_n \rightarrow q)$$

Each of the "or" conditions in the premise is a separate case that must be proven.



Proof by Cases

Example 3.7: Proof by Cases

Proposition 3.8

Let x and y be real numbers. If $x \neq 0$ or $y \neq 0$ then $x^2 + y^2 \neq 0$

Proof by Cases.

First, Consider $x \neq 0$. Then, $x^2 \neq 0$ and $x^2 > 0$. We also know $y^2 \geq 0$ without any further assumptions. $\therefore x^2 + y^2$ is strictly greater than 0

Second, consider $y \neq 0$. Again, $y^2 \neq 0$ and $y^2 > 0$. $\therefore$ since $x^2 \geq 0$, $x^2 + y^2$ is strictly greater than 0 □

Notice in the proof for proposition 3.8 each of the case is identical



Without loss of generality - W.L.O.G.

This a way for provers to be lazy. When two cases are sufficiently similar, one only needs to give a proof to one of the cases.

What constitutes "sufficiently similar" ?

Typically, it will be an arbitrary choice made throughtout the proof for which the other choice would be equally as valid.

Such an arbitrary choice should be symmetric.



W.L.O.G.

Example 3.8: W.L.O.G.

Proposition 3.9

For two integers x and y if xy and $x + y$ are both even, then both x and y are even

Proof.

Prove by contrapositive: Suppose it is not the case that x and y are even. We look to prove xy and $x + y$ are both not even. We prove this by cases

WLOG, assume x is odd. Thus $x = 2k + 1$ for some integer k

Case 1: y is even and $y = 2\ell$ for some ℓ

$$x + y = (2k + 1) + 2\ell = 2(k + \ell) + 1$$

$\therefore x + y$ is odd

Case 2: y is odd and $y = 2\ell + 1$ for some integer ℓ

$$xy = (2k + 1)(2\ell + 1) = 4k\ell + 2k + 2\ell + 1 = 2(2k\ell + k + \ell) + 1$$

$\therefore xy$ is odd

In either case at least one of $x + y$ or xy is odd. This proves the contrapositive and thus the proposition □



W.L.O.G.

In 3.8 we used w.l.o.g. to state that we made an arbitrary choice. We chose that x is odd.

We could have equally made the choice that y is odd and then proved the two cases.

Since either choice would follow the same proof, they are "sufficiently similar" hence need to only prove one of the choices.



Disproof by Counterexample

We have observed that many theorems are stated in the form $\forall x P(x)$.

Therefore, rather than prove such a statement, the statement can be disproved. In other words, one can prove that $\forall x P(x)$ is false.

From Predicate logic, we know that if $\forall x P(x)$ is false, then $\neg \forall x P(x)$

Moreover

$$\neg \forall x P(x) \equiv \exists x \neg P(x)$$

A *proof by example* is a proof $\exists x Q(x)$ by finding a particular example for which $Q(x)$ is true.

Letting $Q(x)$ be $\neg P(x)$ we can thus disprove our original statement $\forall x P(x)$



Disproof by Counterexample

Example 3.9: Disproof by Counterexample

Conjecture 3.2

For a real number x if $x < 1$, then $x^2 < 1$

Proof.

Let $P(x) := x < 1$. Let $Q(x) := x^2 < 1$

We look to disprove $\forall x (P(x) \rightarrow Q(x))$

That is, we should prove $\exists x \neg(P(x) \rightarrow Q(x))$

This implication is false if $P(x)$ is true and $Q(x)$ is false.

$x = -2$ is such a counterexample. $P(-2)$ is true since $-2 < 1$

However, $Q(-2)$ is false since $(-2)^2 = 4 \not< 1$

$\therefore$ the conjecture is *false*





Other than the above proves: Direct Proof and Indirect Proofs, we now explore other proof concepts:

- ① Existence of Proof
- ② Uniqueness Proofs



Existence of Proof

This look to prove theorems of the form $\exists x P(x)$. This is very similar to using counterexamples to disprove a statement.

There are two kinds of existence proofs for $\exists x P(x)$:

- 1 A **Constructive Proof** finds a *witness* a such that $P(a)$ is true. Proving that the statement with the existential quantifier is true. Constructive proofs may provide the witness itself or provide a method for "constructing" a witness.
- 2 A **Nonconstructive Proof** proves that a witness must exist (usually in some indirect way) without providing a particular witness.



Constructive Proof

Example 3.10: Constructive Proof

Theorem 3.6

There is a positive integer that is equal to two different sums of cubes of positive integers

Proof.

$$1729 = 10^3 + 9^3 = 12^3 + 1^3$$



The Challenge with existence theorems is that of finding a witness. Moreover, it is often equally as difficult to disprove an existence theorem.



Open Problems

- Problems that have not yet been proven to be true, but also not yet proved to be false, are open problems.
- Open problems often have a conjecture associated with them. It is often the case that research continues in certain directions with the assumption that a conjecture is true.
- If a conjecture is later found to be false, then all of that work which assumed the opposite is immediately invalidated. Such open problems are therefore very “high stakes”.
- The biggest, and arguably most consequential, open problem is $\mathcal{P} \stackrel{?}{=} \mathcal{NP}$. This question says, can any problem which is easily verified be easily solved?
- Once we have a witness, it is very easy to verify if that witness satisfies the theorem. But, is it always easy to find a witness? That is the open question of whether or not $\mathcal{P} = \mathcal{NP}$
- Most of computer science has been developed under the assumption that $\mathcal{P} \neq \mathcal{NP}$. For example, everything related to cryptography and cybersecurity relies on $\mathcal{P} \neq \mathcal{NP}$



Nonconstructive Proof

Example 3.11: Nonconstructive Proof

Theorem 3.7

There exists irrational numbers x and y such that x^y is rational.

Proof.

We know $\sqrt{2}$ is irrational. Let $x = y = \sqrt{2}$

We have two cases: $\sqrt{2}^{\sqrt{2}}$ is rational or irrational. In this case, the theorem is obviously true and we are done.

Consider the second case. Let $x = \sqrt{2}^{\sqrt{2}}$. Let $y = \sqrt{2}$. By assumption x is irrational and we know y is irrational and we know:

$$x^y = \sqrt{2}^{\sqrt{2} \cdot \sqrt{2}} = \sqrt{2}^{\sqrt{2}\sqrt{2}} = \sqrt{2}^2 = 2$$

Surely 2 is rational, and we have proved the statement in both cases. □



Uniqueness proofs

These are a form of existence proofs where we want to prove a statement of the form “there exists a *unique* x such that $P(x)$ ”

This kind of theorem statement can be written using special notation: $\exists!x P(x)$. $\exists!$ means that there exists exactly one element from the domain of discourse that satisfies the predicate.

Proving uniqueness can be very tricky. Indeed, proving $\exists!x P(x)$ is the same as proving a more complicated statement:

$$\exists x (P(x) \wedge \forall y (y \neq x \rightarrow \neg P(y)))$$

or equivalently

$$\exists x (P(x) \wedge \forall y (P(y) \rightarrow y = x))$$



Uniqueness proofs

In this case, there are two parts of a uniqueness proof.

- 1 *Existence* we must show that an x exists such that $P(x)$ is true
- 2 *Uniqueness* we must show if $y \neq x$ then $P(y)$ is *false* or we must show that if $P(y)$ is true then $y = x$ (they are contrapositives).



Example 3.12: Uniqueness proofs

Theorem 3.8

If a and b are real numbers such that $a \neq 0$, then there is a unique real number r such that $ar + b = 0$

Proof.

Existence: Since $a \neq 0$ we have $ar + b = 0 \implies r = \frac{-b}{a}$

Uniqueness: Proceed by contradiction.

Suppose s is a real number such that $s \neq r$ and $as + b = 0$ Since this and the previous equation both equal 0 we have:

$$ar + b = as + b$$

$$ar = as$$

$$r = s$$

Since we chose s arbitrarily and arrived at the fact the $r = s$ we know that r is unique. □



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Sixth Slide



Sixth Slide



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